Samrudha Narendra

Bengaluru, India
+91 9731514426 | samrudhnarendra5@gmail.com | LinkedIn | GitHub

Education

PES University 2023 – 2027
B.Tech in Computer Science Engineering — CGPA: 8.25/10 *Bengaluru, India*

National Public School, Rajajinagar 2019 – 2023
Senior High School — Percentage: 91.2%

Experience

Summer Research Intern June 2025 – July 2025
CCBD-CDASML Research Centre, PES University *Bengaluru, India*

- Fine-tuned Whisper ASR models on Kannada speech datasets with phoneme-level annotations using PyTorch
- Evaluated Automatic Speech Recognition performance using WER/CER metrics across gender-specific datasets
- Analyzed dataset variability and alignment constraints affecting model accuracy
- Published research work at IEEE I3CTCON 2026 [Paper]

Teaching Assistant – C Programming Feb 2026 – Present
PES University *Bengaluru, India*

- Mentored 65+ undergraduate students in C programming, debugging, memory management, and problem-solving through weekly lab sessions.
- Assisted in evaluating assignments and resolving coding issues, helping improve student understanding of pointers, data structures, and file handling concepts.

Projects

RideSync – Secure Campus Ride Sharing Platform | *Java, Spring Boot, JWT, JavaFX, Hibernate, JPA*

- Developed a full-stack campus ride-sharing platform using Spring Boot, JavaFX, Hibernate, and MySQL with role-based workflows for riders, drivers, and admins.
- Secured 40+ REST APIs using JWT authentication and Spring Security; implemented ride booking, OTP verification, and banned-user enforcement.
- Built a layered MVC architecture with JPA repositories and route optimization using Dijkstra's algorithm, supporting ride history, ratings, and audit logging.

Self-Healing Agentic RAG System | Python, LangChain, FAISS, BM25

- Built a Retrieval-Augmented Generation pipeline using hybrid retrieval with BM25 and FAISS
- Implemented hallucination detection, query reformulation, and self-correction workflows
- Improved answer reliability through iterative re-retrieval on HotpotQA datasets

Distributed Image Processing Pipeline | Kafka, Python, Distributed Systems

- Built a distributed image processing pipeline using Kafka and Python, enabling parallel processing of 512 X 512 image tiles across multiple worker nodes.
- Designed heartbeat monitoring and asynchronous task scheduling mechanisms that improved fault tolerance and worker recovery.
- Leveraged Kafka consumer groups for workload balancing, achieving faster processing than a sequential implementation.

Technical Skills

Languages: Python, C, JavaScript, SQL, Java
Frameworks & Libraries: React.js, Node.js, Express.js, PyTorch, LangChain
Machine Learning / AI: NLP, Whisper, Transformers, RAG, FAISS, BM25
Databases: MongoDB, MySQL
Tools & Technologies: Git, GitHub, Kafka, Docker, Postman, Linux, REST APIs
Core CS Subjects: Data Structures and Algorithms, Operating Systems, DBMS, Computer Networks, OOP

Leadership & Activities

Club Head - GronIT Club, PES University 2025 – 2026

- Organized ideathons and state-level hackathons involving 100+ participants
- Managed outreach, sponsorships, logistics, and operations across a 60-member team

Event Management Core - Aikya Club, PES University 2025 – 2026

- Led planning and execution of multiple community and technical events
- Coordinated teams of 10–20 members for logistics and event operations